NEURAL RADIANCE FIELDS (NeRF) IN HOME AND INTERIOR DESIGN

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Outline

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Introduction

1. **What is NeRF?**
   - **Continuous 3D Scene Encoding:** Deep learning framework synthesizing novel 3D views from sparse 2D images.
   - **5D Coordinate Mapping:** Fully connected MLP mapping coordinates (x, y, z) and ray angles (θ, ϕ) to RGB color and volume density (σ).

2. **Background and Context**
   - **Legacy Bottlenecks:** Manual CAD modeling, costly LiDAR rigs, and photogrammetry surface mesh artifacts.
   - **Volumetric Shift:** Direct optimization of radiance fields eliminating traditional polygon mesh creation.

Introduction (Contd.)

3. Scope and Relevance

- **Target Applications:** Interactive real estate walkthroughs, architectural digital twins, and automated virtual staging.
- **Key Capabilities Analyzed:**
  - Bounded indoor scene volumetric parameterization.
  - Single-view and floorplan-guided 3D generative lifting.
  - Object-level furniture decomposition, editing, and relocation.
  - Live monocular SLAM tracking and dense mapping.
- **Core Impact:** Accelerates Spatial AI, AR/VR graphics, and automated interior design workflows.

Motivation

1. **Importance of the Topic**
   - **Industry Demand:** Rapid photorealistic room virtualization for interior design and e-commerce.
   - **Complex Surface Handling:** Native representation of non-Lambertian reflections, mirrors, and intricate textiles.

2. **Current Technical Bottlenecks**
   - **Indoor Scanning Challenges:** Large textureless walls, dim corners, and uneven lighting exposures.
   - **Monolithic Representation:** Inability to isolate, move, or swap individual furniture items in static fields.
   - **Computational Overhead:** Long per-scene training times hindering interactive design workflows.

Motivation (Contd.)

3. Potential Impact of Solving the Problem

- **Casual Capture:** Reconstruct complete 3D rooms from smartphone cameras or 2D blueprints.
- **Interactive Virtual Staging:** Relocate or swap furniture with automatic background surface completion.
- **Physical Realism:** Enforce strict planar constraints to eliminate floating or penetrating furniture artifacts.
- **Production Efficiency:** Real-time rendering output ready for VFX pipelines, gaming engines, and AR headsets.

Basic Terminologies

Volumetric Field: Continuous 3D density and color representation omitting explicit meshes.

Ray Marching: Differentiable sampling along optical rays to numerically integrate pixel colors.

3D Gaussian Splatting: Explicit anisotropic ellipsoids enabling 700+ fps rasterization[cite: 3].

4-DoF Constraint: Restricting furniture motion to estimated 2D floor planes (translation, yaw, scale)[cite: 1].

DeCaPE: Depth-enhanced Camera Positional Encoding for cross-room multi-view alignment[cite: 2].

Occluded Completion: Inferring hidden geometry behind furniture and beneath floorboards[cite: 4].

Classification of NeRF Indoor Architectural Frameworks

Literature Review

Paper 1: 3D-RE-GEN: 3D Reconstruction of Indoor Scenes (Sautter et al.)

- **Single-Image Decomposition:** Extracts clean background meshes and isolated 3D furniture models from one photograph[cite: 1].
- **Application-Querying (A-Q):** GUI-style visual prompting for context-aware inpainting of occluded assets[cite: 1].
- **4-DoF Planar Optimization:** Enforces strict floor-plane alignment via RANSAC ground extraction[cite: 1].
- **Modular Asset Creation:** Couples Hunyuan3D-2.0 and VGGT for camera and asset estimation[cite: 1].

Literature Review (Contd.)

Paper 1: 3D-RE-GEN (Contd.)

Summary of Research Gaps

- **Segmentation Error Cascading:** High failure rates if Grounded-SAM produces faulty initial boundary masks[cite: 1].
- **Non-Planar Asset Failure:** Lack of hierarchical constraints for secondary objects (e.g., cups on tables, books on shelves)[cite: 1].
- **High Compute Overhead:** 17–20 min execution per scene on a single RTX 4090 GPU[cite: 1].

Literature Review (Contd.)

Paper 1: 3D-RE-GEN (Architecture)

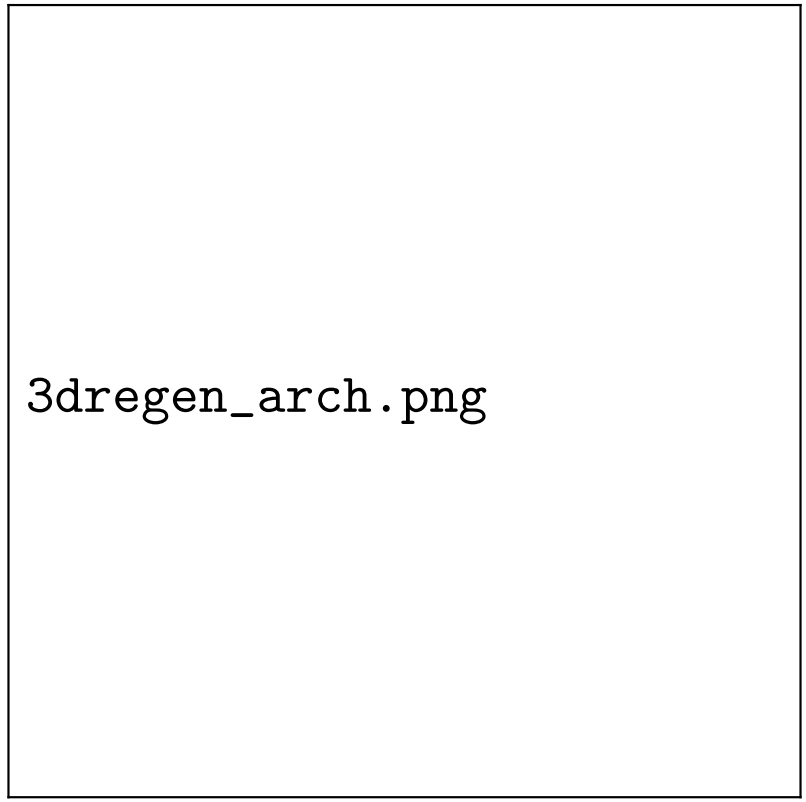


Fig 1.1: 3D-RE-GEN Modular Decomposition Framework[cite: 1]

Literature Review (Contd.)

Paper 2: HouseCrafter: Lifting Floorplans to 3D Scenes (Chen et al., ICCV 2025)

- **2D-to-3D Lifting:** Converts vectorized architectural floorplans directly into navigable multi-room 3D houses[cite: 2].
- **Autoregressive Batch Synthesis:** Generates sequential multi-view RGB-D images across camera location graphs[cite: 2].
- **DeCaPE Encoding:** Depth-enhanced Camera Positional Encoding preserving geometric cross-room alignment[cite: 2].
- **Explicit Surface Fusion:** Direct TSDF integration into clean, textured polygonal meshes[cite: 2].

Literature Review (Contd.)

Paper 2: HouseCrafter (Contd.)

Summary of Research Gaps

- **Drift Accumulation:** Error propagation across long autoregressive camera paths in large houses[cite: 2].
- **Monolithic Meshing:** Produces unified geometry without object-level semantic segmentation or editing[cite: 2].
- **Layout Ambiguity:** Performance degradation on irregular, non-orthogonal architectural floorplans[cite: 2].

Literature Review (Contd.)

Paper 2: HouseCrafter (Architecture)

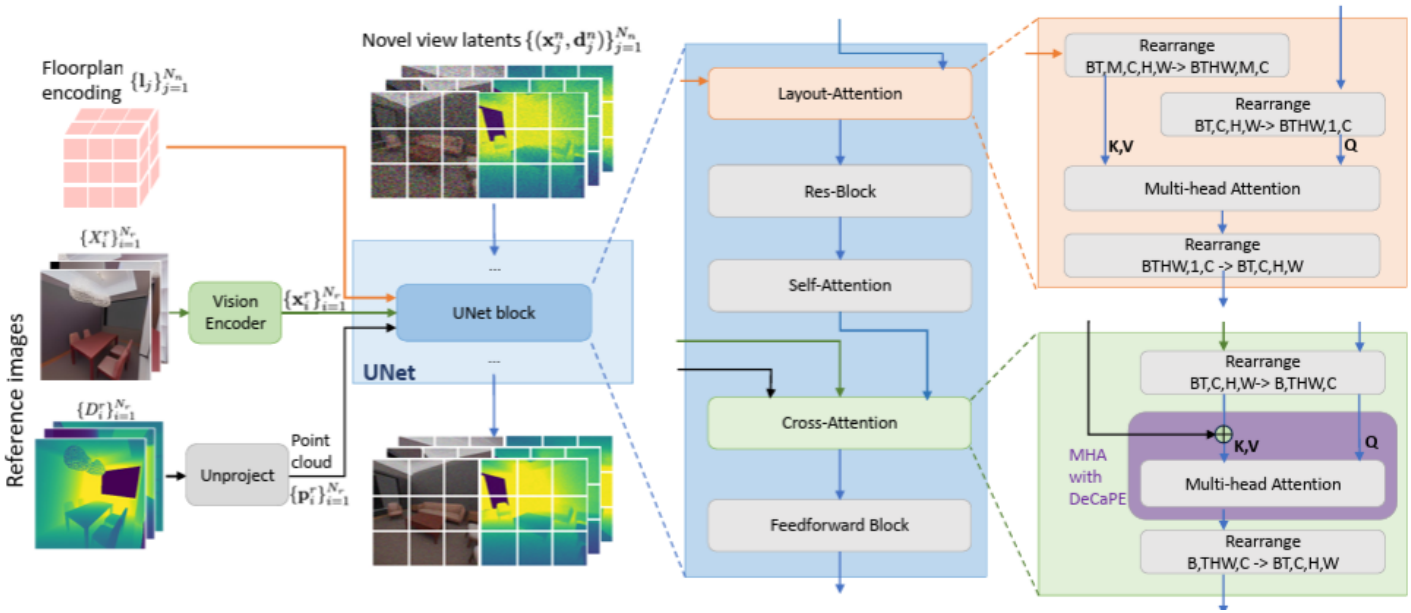


Fig 2.1: Floorplan-Guided Multi-View RGB-D Diffusion Architecture[cite: 2]

Literature Review (Contd.)

Paper 3: Gaussian Splatting SLAM (Matsuki et al., CVPR 2024)

- **Unified Map Representation:** First online monocular/RGB-D SLAM system relying entirely on 3D Gaussian Splatting[cite: 3].
- **Direct Analytical Tracking:** SE(3) Lie algebra Jacobians for real-time camera tracking directly against Gaussians[cite: 3].
- **Isotropic Shape Regularization:** Constrains Gaussian elongation along viewing rays to eliminate line-of-sight artifacts[cite: 3].
- **Live Performance:** Achieves 3 fps tracking and interactive 700+ fps rendering speeds[cite: 3].

Literature Review (Contd.)

Paper 3: Gaussian Splatting SLAM (Contd.)

Summary of Research Gaps

- **Missing Global Loop Closure:** Susceptible to drift over large multi-room trajectory paths[cite: 3].
- **Non-Surface Geometry:** Absence of explicit boundary meshes complicates CAD export and collision physics[cite: 3].
- **VRAM Footprint:** Dynamic allocation requires substantial GPU memory for extended exploration sessions[cite: 3].

Literature Review (Contd.)

Paper 3: Gaussian Splatting SLAM (Architecture)

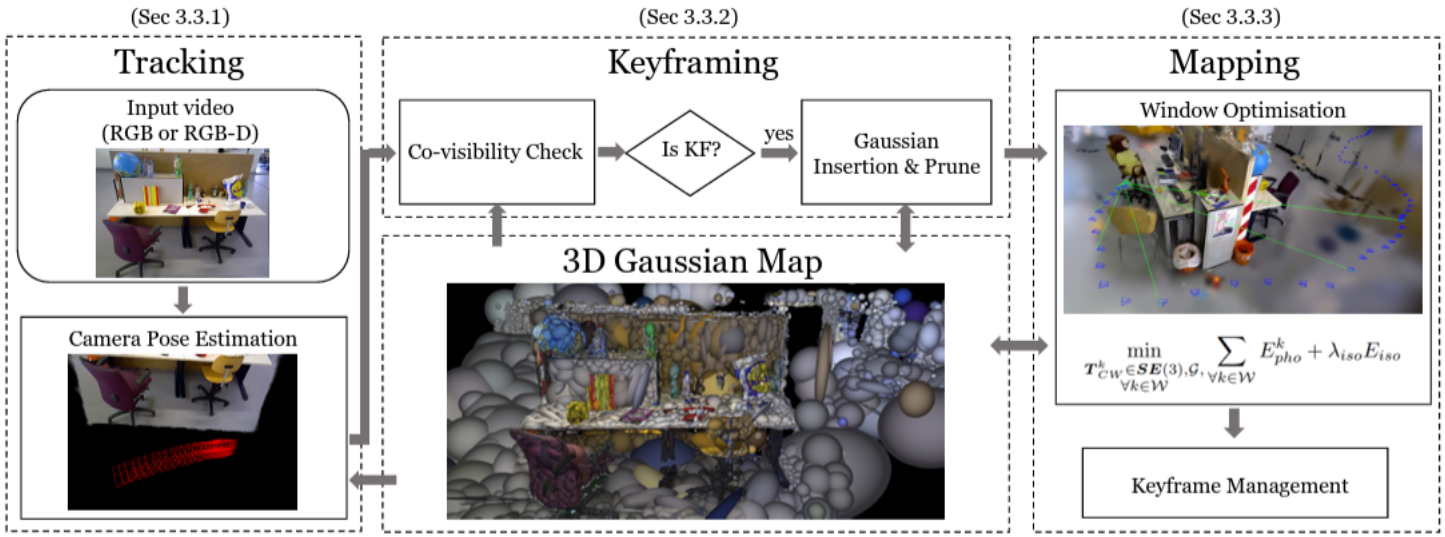


Fig 3.1: 3DGS Direct Tracking, Keyframing, and Mapping Pipeline[cite: 3]

Literature Review (Contd.)

Paper 4: Behind the Veil: Occluded Surfaces Completion (Sun et al., CVPR 2024)

- **Invisible Surface Inpainting:** Completes hidden areas (floors under sofas, wall boundaries behind cabinets)[cite: 4].
- **Hierarchical Octree Structure:** Coarse octree features represent scene context; fine features encode visible details[cite: 4].
- **Dual-Decoder Scheme:** Online-optimized Geo-decoder paired with a cross-scene pre-trained 3D Inpainter[cite: 4].
- **Benchmark Superiority:** Outperforms SOTA by up to 24.2% in surface completeness metrics[cite: 4].

Literature Review (Contd.)

Paper 4: Behind the Veil (Contd.)

Summary of Research Gaps

- **Sensor Dependency:** Strictly requires structured depth readings; incapable of RGB-only operation[cite: 4].
- **Geometric Hallucination:** Shape distortion when completing unseen, highly customized furniture assets[cite: 4].
- **Static Assumption:** Cannot adapt to dynamic indoor scenes where objects move during scanning[cite: 4].

Literature Review (Contd.)

Paper 4: Behind the Veil (Architecture)

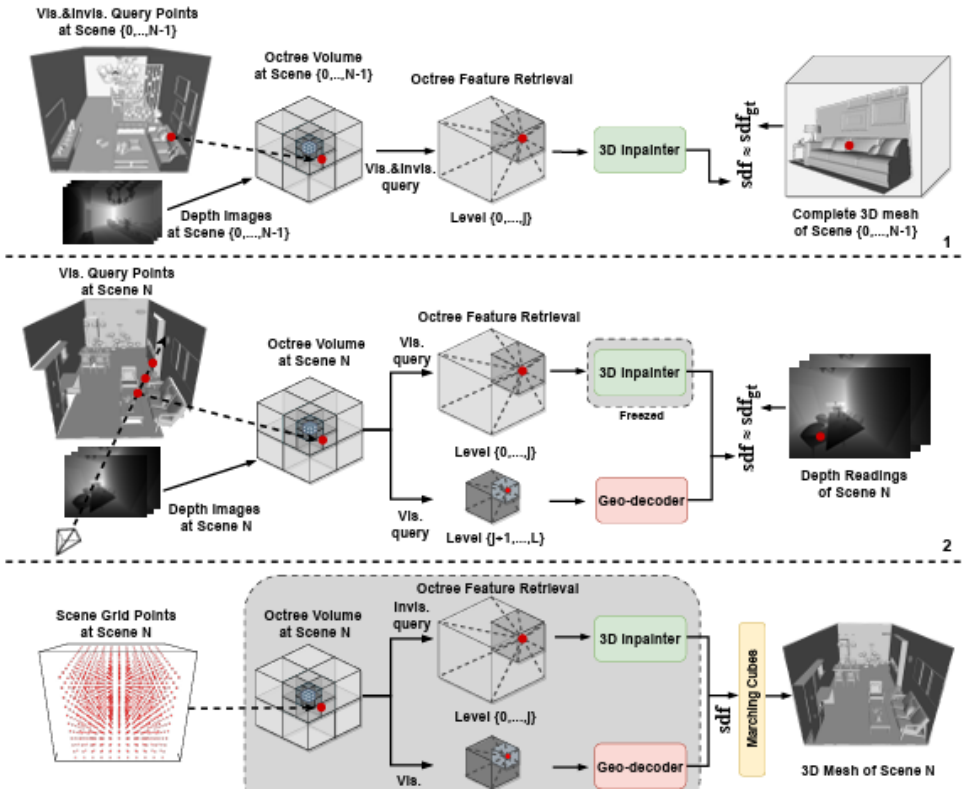


Fig 4.1: Dual-Decoder Octree Architecture for Occluded Surface Completion[cite: 4]

Comparison Report

Table 1: Indoor Scene Reconstruction Approaches Comparison

Paper Name	Core Methodology	Key Advantages	Primary Limitations
3D-RE-GEN (Sautter et al.)	4-DoF planar constraint + GUI Application-Querying[cite: 1]	Disentangles background and modular 3D furniture assets[cite: 1]	Mask error sensitivity; heavy diffusion backend[cite: 1]
HouseCrafter (Chen et al.)	Autoregressive RGB-D diffusion + DeCaPE[cite: 2]	Generates complete house-scale scenes from 2D floorplans[cite: 2]	Cumulative batch drift; non-editable static mesh[cite: 2]
Gaussian SLAM (Matsuki et al.)	3DGS + SE(3) Lie algebra tracking[cite: 3]	Live 3 fps tracking; renders thin structures and transparency[cite: 3]	Lacks global loop closure; no explicit mesh output[cite: 3]
Behind Veil (Sun et al.)	Coarse-fine octree + dual-decoder inpainting[cite: 4]	Completes obscured floors, walls, and furniture backsides[cite: 4]	Requires depth sensor input; custom shape errors[cite: 4]

Conclusion

- **Technological Leap:** Shifted indoor spatial computing from manual CAD pipelines to continuous neural representations.
- **Physical Plausibility:** 4-DoF planar constraints successfully eliminate floating and intersecting asset errors[cite: 1].
- **Generative Architectural Lifting:** 2D floorplans now reliably bootstrap consistent multi-room 3D environments[cite: 2].
- **Real-Time Immersion:** 3D Gaussian Splatting provides live, interactive tracking without expensive ray marching[cite: 3].
- **True Spatial Completeness:** Dual-decoder inpainting eliminates hidden holes, providing production-ready virtual staging[cite: 4].

References

[1] T. Sautter, J.-N. Dihlmann, and H. Lensch, "3D-RE-GEN: 3D Reconstruction of Indoor Scenes with a Generative Framework," *arXiv preprint arXiv:2512.17459*, Dec. 2025[cite: 1].

[2] Y. Chen, H. T. Nguyen, V. Voleti, V. Jampani, and H. Jiang, "HouseCrafter: Lifting Floorplans to 3D Scenes with 2D Diffusion Models," *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025[cite: 2].

[3] H. Matsuki, R. Murai, P. H. J. Kelly, and A. J. Davison, "Gaussian Splatting SLAM," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 18039–18048, 2024[cite: 3].

[4] S. Sun, C. Zhao, Y. Guo, R. Wang, X. Huang, Y. V. Chen, and L. Ren, "Behind the Veil: Enhanced Indoor 3D Scene Reconstruction with Occluded Surfaces Completion," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 12744–12753, 2024[cite: 4].

THANK YOU

Any Questions?

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